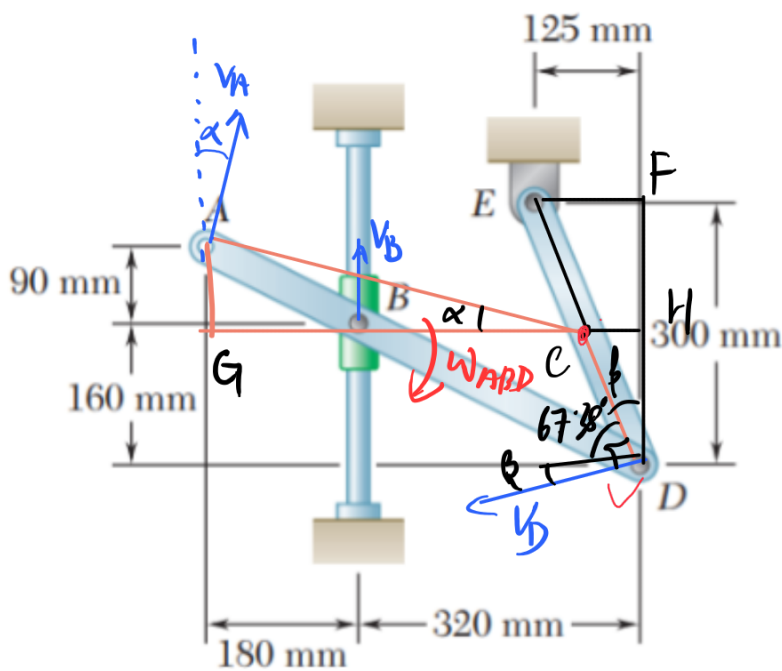


15.90 Arm ABD is connected by pins to a collar at B and to crank DE . Knowing that the velocity of collar B is 400 mm/s upward, determine (a) the angular velocity of arm ABD , (b) the velocity of point A .

Grade: 414 . VG!



C is the instantaneous center of zero velocity of bar ABD .
very close to $\vec{V}_B$, $\vec{V}_D$ directions

$$V_B = 400 \text{ mm/s} \uparrow$$

$$\tan \beta = \frac{ED}{FD} = \frac{125}{300} \quad \beta = 22.62^\circ$$

(a) $BC + CH = 320 \text{ mm}$ ✓

$$(CH = HD) \tan \beta = 160 \left(\frac{125}{300} \right) = \frac{200}{3} \quad \checkmark$$

$$BC = 320 \text{ mm} - CH \quad \checkmark$$

$$= \left(320 - \frac{200}{3} \right) \text{ mm}$$

$$= \frac{760}{3} \text{ mm}$$

$$V_B = \omega_{ABD} \cdot r_{B/IC} = \omega_{ABD} \cdot BC$$

$$400 = \omega_{ABD} \cdot \frac{760}{3}$$

$$\omega_{ABD} = 1.579 \text{ rad/s} \downarrow \quad \checkmark$$

(b) $GC + CH = 180 + 320$ ✓

$$\alpha = \tan^{-1} \left(\frac{AG}{GV} \right) = 11.73^\circ$$

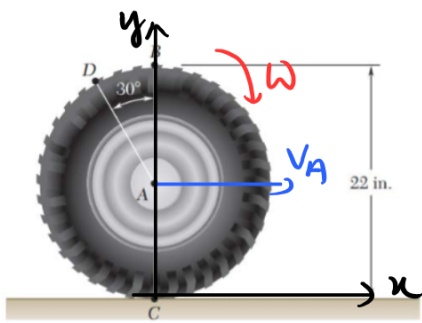
$$GC = 180 + 320 - \frac{200}{3} = \frac{1300}{3} \text{ mm}$$

$$AC = \frac{GC}{\cos \alpha}$$

$$AC = 442.58 \text{ mm}$$

$$V_A = \omega_{ABD} \cdot r_{A/IC}$$

$$V_A = 1.579 (AC) = 698.83 \text{ mm/s} \quad \angle 78.27^\circ$$



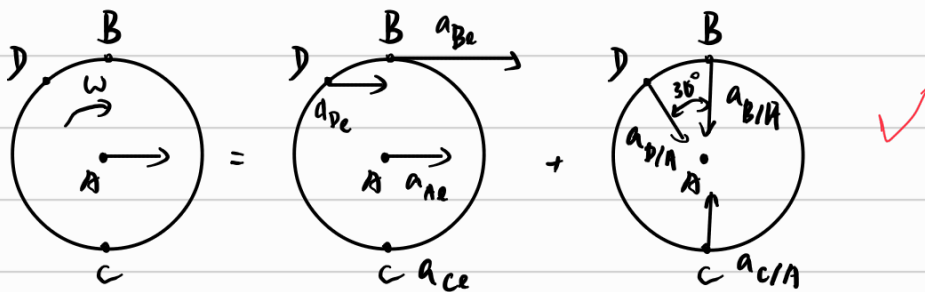
15.111 An automobile travels to the left at a constant speed of 48 mi/h. Knowing that the diameter of the wheel is 22 in., determine the acceleration (a) of point B, (b) of point C, (c) of point D.

$$V_A = 48 \frac{\text{mi}}{\text{h}} \cdot \frac{1}{3600 \text{ s}} \cdot \frac{63360 \text{ in}}{\text{mi}} = 844.8 \text{ in/s}$$

$$V_C = V_A + \omega \times r_{AC}$$

$$0 = V_A i - R \omega i$$

$$\omega = \frac{V_A}{R} = \frac{844.8}{11} = 76.8 \text{ rad/s}$$



$$a_B = a_A + a_{B/A}$$

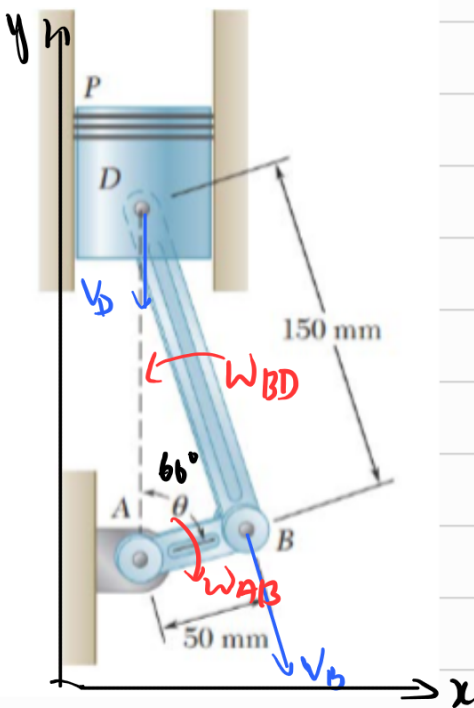
$$a_{B/A} = a_{C/A} = a_{D/A} = r\omega^2 = 11(76.8)^2 = 64880.64 \text{ in/s}^2$$

(a) $a_B = 0 + 64880.64 \text{ in/s}^2 \downarrow$

(b) $a_C = 64880.64 \text{ in/s}^2 \uparrow$

(c) $a_D = 64880.64 \text{ in/s}^2 \nearrow 60^\circ$

15.125 Knowing that crank AB rotates about point A with a constant angular velocity of 900 rpm clockwise, determine the acceleration of the piston P when $\theta = 60^\circ$.



$$\omega_{AB} = 900 \text{ rpm} = \frac{900}{60} \times 2\pi = 30\pi \text{ rad/s} \quad \checkmark$$

$$V_B = \omega_{AB} \cdot r_{AB} = 30\pi \left(\frac{50}{1000}\right) = 1.5\pi \text{ m/s} \quad \angle 60^\circ \quad \checkmark$$

$$V_D = V_D \downarrow \quad \omega_{BD} = \omega_{BD} \quad \checkmark$$

$$V_{D/B} = 0.15 \omega_{BD} \quad \angle \beta \quad \checkmark$$

$$\begin{aligned} & \text{Velocity triangles: } \triangle V_B, \triangle V_{D/B}, \triangle V_D \\ & \frac{\sin \beta}{0.05} = \frac{\sin 60}{0.15} \\ & \beta = \sin^{-1} \left(0.05 \frac{\sin 60}{0.15} \right) = 16.78^\circ \quad \checkmark \end{aligned}$$

$$[V_D \downarrow] = [V_B \angle 60^\circ] + [V_{D/B} \angle \beta] \quad \checkmark$$

components in x direction = 0

$$V_B \cos 60^\circ - V_{D/B} \cos \beta = 0 \quad \checkmark$$

$$1.5\pi \cos 60^\circ - 0.15 \omega_{BD} \cos \beta = 0 \quad \checkmark$$

$$0.75\pi - 0.15 \omega_{BD} \cos \beta = 0 \quad \checkmark$$

$$\omega_{BD} = \frac{0.75\pi}{0.15 \cos \beta} = 16.4065 \text{ rad/s} \quad \checkmark$$

$$a_D = a_B + a_{D/B}$$

$$\epsilon_{AB} = 0 \quad \checkmark$$

$$a_B = r\omega^2 = 0.05 \omega_{AB}^2$$

$$= 0.05 (30\pi)^2 = 444.13 \text{ m/s}^2 \quad \angle 30^\circ \quad \checkmark$$

$$a_{D/B} = [0.15 \alpha_{BD} \angle \beta] + [0.15 \omega_{BD}^2 \angle \beta] \quad \checkmark$$

$$= [6 \alpha_{BD} \angle \beta] + [40.376 \angle \beta]$$

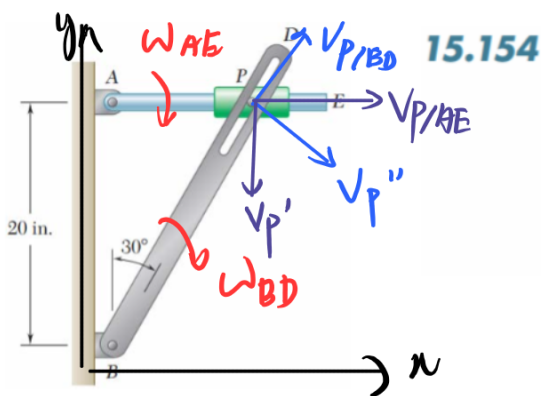
$$a_D = a_B + a_{D/B}$$

$$\text{components on x direction: } 0 = -444.13 \cos 30^\circ + 0.15 \alpha_{BD} \cos \beta + 40.376 \sin \beta$$

$$\alpha_{BD} = 2597 \text{ rad/s}^2$$

$$\begin{aligned} \text{components on y direction: } a_D &= 444.13 \sin 30^\circ - 0.15 (2597) \sin \beta + 40.376 \cos \beta \\ &= 148.27 \text{ m/s}^2 \end{aligned}$$

$$a_D = a_D = 148.27 \text{ m/s}^2 \quad \downarrow$$

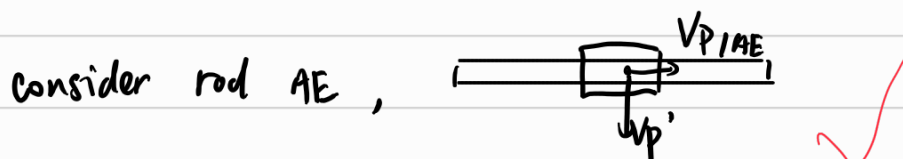


Pin P is attached to the collar shown; the motion of the pin is guided by a slot cut in rod BD and by the collar that slides on rod AE. Knowing that at the instant considered the rods rotate clockwise with constant angular velocities, determine for the given data the velocity of pin F.

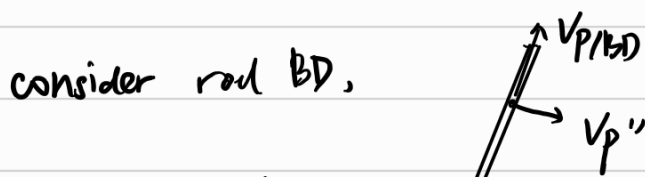
15.154 $\omega_{AE} = 4 \text{ rad/s}$, $\omega_{BD} = 1.5 \text{ rad/s}$.

$$AP = 20 \tan 30^\circ = 11.547 \text{ in}$$

$$BP = \frac{20}{\cos 30^\circ} = 23.094$$



$$\vec{V}_P = \vec{V}_P' + \vec{V}_{P/AE} \quad (1)$$



$$\vec{V}_P = \vec{V}_P'' + \vec{V}_{P/BD} \quad (2)$$

Along x-axis:

$$V_{P/AE} = V_P'' \cos 30^\circ + V_{P/BD} \sin 30^\circ$$

$$V_{P/AE} = 23.094(1.5) \cos 30^\circ + \frac{1}{2} V_{P/BD}$$

$$V_{P/AE} = 23.0 + \frac{1}{2} (33.3333)$$

$$= 13.3334 \text{ in/s}$$

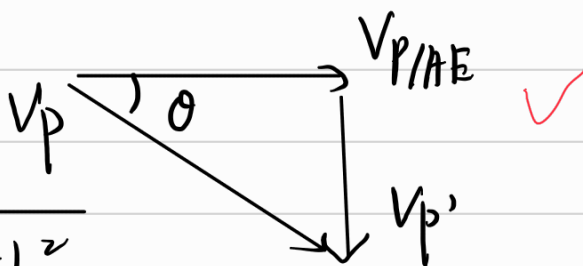
Along y-axis:

$$V_P' = V_P'' \sin 30^\circ - V_{P/BD} \cos 30^\circ$$

$$11.547(4) = 23.094(1.5) \left(\frac{1}{2}\right) - V_{P/BD} \cos 30^\circ$$

$$V_{P/BD} = -33.3333 \text{ in/s}$$

$$\vec{V}_P = \vec{V}_P' + \vec{V}_{P/AE}$$



$$V_P = \sqrt{(46.188)^2 + (13.3334)^2}$$

$$= 48.074 \text{ in/s}$$

$$\tan \theta = \frac{v_p'}{v_{p/AE}} \quad \theta = \tan^{-1} \left(\frac{v_p'}{v_{p/AE}} \right) = 73.9^\circ \quad \checkmark$$

$$v_p = 48.074 \quad \searrow 73.9^\circ \quad \checkmark$$